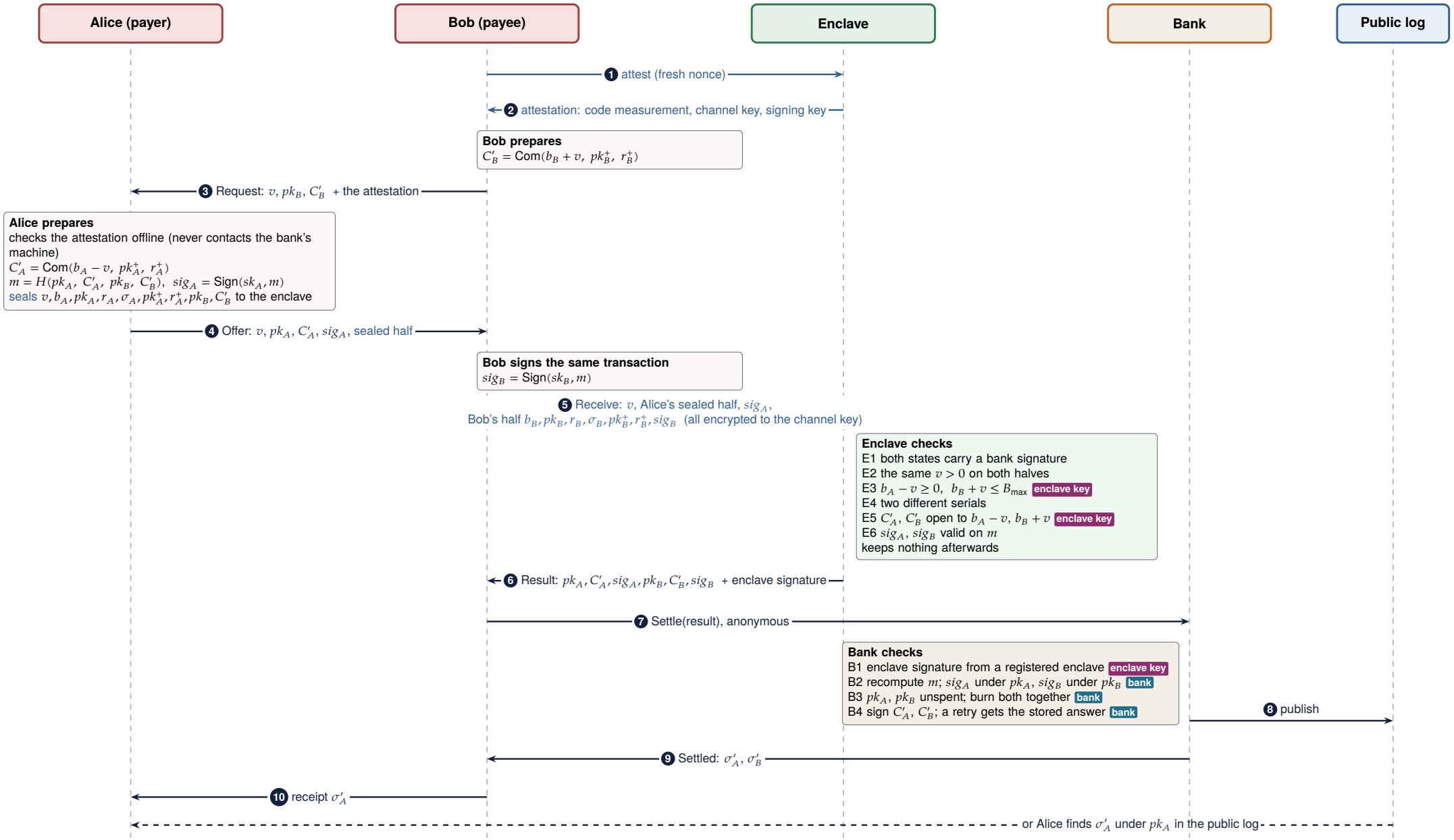
Round 1: a payment

Alice pays v to Bob. As in Platypus, only the recipient talks to the system: Bob alone contacts the enclave and the bank

blue sealed to the attested enclave **bank** check the bank does itself **enclave key** rests on the enclave's signing key
$^+$ next value, for example pk_A^+ is Alice's next serial key $\mathrm{Com}(b, pk, r)$ hides a state, σ is the bank's signature on it



Why both owners sign $m = H(pk_A, C'_A, pk_B, C'_B)$**.** Each serial is a one-time public key, so only the owner of a state can spend it, and the bank checks this itself (B2): signing keys never reach the enclave. The two signed commitments fix both new balances, so the amount and the payee cannot be changed. v is left out of m so the bank can recompute m without learning the amount. **What stays on the enclave's key:** balance conservation (E3, E5), which the bank cannot see.